TOPS Technologies

Interview Questions — PYTHON FOR DATA ANALYTICS

Total Questions: 63

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1. What is Python and why is it commonly used in data analytics?

> **Hint:** Think about Python's features and its popularity in handling data.

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2. How would you install Python and set up Jupyter Notebook for your data analytics work?

> **Hint:** Consider tools like Anaconda and why Jupyter is helpful for analysts.

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3. What is the difference between a .py file and a .ipynb file in Python?

> **Hint:** One is a script, the other is a notebook format.

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4. Write a Python statement that prints 'Hello Analyst' in a Jupyter Notebook cell.

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5. Why is indentation important in Python?

> **Hint:** Think about how Python defines code blocks.

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6. What are the rules for naming variables in Python?

> **Hint:** Consider allowed characters and good practices.

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7. How do you write a single-line comment and a multiline comment in Python?

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8. Create three variables in Python to store sales, region, and discount values.

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9. List the four main built-in data types in Python that are important for analytics.

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10. How does dynamic typing work in Python? Give an example.

> **Hint:** Think about assigning different types of values to the same variable.

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11. How would you convert a string like '123.45' to a float in Python?

> **Hint:** Recall the type casting functions.

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12. What is the difference between an int and a float in Python?

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13. Name three common string methods in Python and briefly describe what they do.

> **Hint:** Think about changing case, replacing text, or splitting.

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14. How would you remove extra spaces and convert a product name like ' samsung-Mobile ' to 'Samsung-Mobile' in Python?

> **Hint:** Consider using strip() and case conversion methods.

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15. Show how to extract the first five characters from a string variable called product_name.

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16. Explain the difference between a list and a tuple in Python. When would you use each?

> **Hint:** Consider mutability and use cases.

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17. Write Python code to create a list of customer IDs, add a new ID to it, and then remove the last ID.

> **Hint:** Use append() and pop().

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18. How would you iterate over a list of sales amounts and print each value?

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19. What is a dictionary in Python? Give an example relevant to data analytics.

> **Hint:** Think of key-value pairs, like product-price.

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20. How can you add a new key-value pair to a dictionary and update the value of an existing key?

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21. Write a Python set operation to find common customer IDs between two sets.

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22. How does an if-elif-else statement work in Python? Give an example with revenue and target.

> **Hint:** Think about decision-making based on values.

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23. Write a Python code snippet that prints 'Achieved' if revenue is greater than target, otherwise prints 'Not Achieved'.

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24. What is the difference between a for loop and a while loop in Python?

> **Hint:** Consider when each is used.

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25. Write a for loop to calculate the total sales from a list called sales_list.

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26. How does the break statement affect loop execution in Python?

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27. Define a function in Python called calculate_discount that takes price and rate as arguments and returns the discounted price.

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28. What are default arguments in Python functions? Provide an example.

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29. Explain the difference between a regular function and a lambda function in Python.

> **Hint:** Consider syntax and typical use cases.

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30. How would you use map() to apply the strip() method to clean all product names in a list called products?

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31. Write a Python statement using reduce() to calculate the total of a sales list.

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32. What is recursion in Python? Give an example where it might be useful.

> **Hint:** Think about functions calling themselves.

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33. How do local and global variables differ in their scope within a Python program?

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34. Write a recursive function to calculate the factorial of a number n.

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35. How would you read a CSV file in Python and print each row?

> **Hint:** Remember the csv module.

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36. What is the difference between reading a JSON file and a text file in Python?

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37. Write Python code to open a text file called 'data.txt', read its contents, and print them.

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38. How does exception handling work in Python? Why is it important for data analytics?

> **Hint:** Think about try-except blocks and avoiding crashes.

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39. Write a Python function that divides two numbers and handles the ZeroDivisionError gracefully.

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40. What is the purpose of the finally block in exception handling?

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41. Explain the difference between an iterator and a generator in Python.

> **Hint:** Consider how values are produced and memory usage.

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42. Write a simple generator function that yields sales values one by one from a list.

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43. How would you use the zip() function to combine two lists: product_names and product_prices?

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44. How do you import a built-in module like math in Python and use its sqrt function?

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45. What is the purpose of a virtual environment (venv) in Python projects?

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46. Describe the steps to create and use a custom module called utils.py in your data analytics project.

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47. How would you use pip to install an external package such as pandas?

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48. What is a regular expression and how can it help in cleaning data in Python?

> **Hint:** Think about pattern matching.

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49. Write a Python statement using re.findall() to extract all email addresses from a string variable called text.

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50. How would you use re.sub() to remove all digits from a string?

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51. What is a class in Python? Give a simple example relevant to analytics, such as a Customer class.

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52. How does the __init__() method work in a Python class?

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53. Write a Python class called Customer with attributes name, age, and purchase_history.

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54. What is encapsulation in Python and how do you make an attribute private?

> **Hint:** Consider using underscores.

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55. How do getters and setters help manage access to private attributes in a class?

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56. Explain abstraction in Python OOP with a simple example.

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57. What is inheritance in Python? How would you create a Manager class that inherits from an Employee class?

> **Hint:** Think about the class definition syntax.

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58. How does method overriding work in Python? Give an example with an overridden bonus method.

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59. What is polymorphism in Python and how can it be useful in analytics projects?

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60. How could you use an AI tool like ChatGPT or GitHub Copilot to help you debug a Python script for data analysis, and what steps would you take to verify the AI's suggestions?

> **Hint:** Think about code review, testing, and not trusting AI output blindly.

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61. If you are cleaning a large dataset in Python, how could AI-based tools speed up your workflow? In what situations would you still rely on your own Python skills instead of AI?

> **Hint:** Consider repetitive tasks vs. understanding data context.

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62. Suppose you have to process customer data from a CSV file, clean product names, and calculate total sales. Which Python features or techniques from this module would you use together to complete this task?

> **Hint:** Think about file handling, string methods, loops, and aggregation.

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63. Imagine you are writing a data analytics script that needs to handle missing data and possible errors in input files. How would you combine exception handling, file operations, and data type checks to make your script robust?
